

Tab 1

Introduction: Food and Friendship in Learning

This excerpt discusses the growth in my understanding of the engineering design process by reflecting on a particular observation through the lens of a particular Praxis CTMF (concept, theory, model, framework). This observation came from working with a team to develop solutions to the issue of struggling to get snacks out of a vending machine - we were trying to improve the overall product retrieval process. I am going to analyze how a supportive, “friend”-like, team environment can affect understanding of concepts and the work produced. This enriched my understanding of the engineering design process and has given me key takeaways to apply in the future.

Observation: A True Team

Having a strong team dynamic creates an optimal space for learning and creating, but may detract from accountability and efficiency. The team I got to work with throughout the course had an incredible bond. As a group of all women in engineering, we shared a lot of values and perspectives and quickly became friends. We often went on socials and enjoyed each other's company (figure 1) However, I observed that this friendship occasionally got in the way of our productivity. Our discussions often got off topic and veered towards personal interests rather than the assignment at hand (figure 2).

Impact: United but Inefficient

This team dynamic impacted our work in various ways. Figures three and four have samples of the ITP Team feedback from early November. The positive feedback primarily focuses on the successes of the team environment we have created; we were able to collaborate effectively, and had a strong shared culture. The group was supportive and we were able to create a safe space to take risks and suggest ideas. . We had fun working developing our assignments and this showed in our work,(figure 5) However, we were slow in completing our work, (figure 6.1 and 6.2). We often handed in our assignments late and had to “cram” to finish it We had prioritized culture over progress and it often created stress as the deadlines neared.

Analysis: The Influence of Comfort and Support while Learning and Designing

I have a new understanding of Vygotskian learning theory and its impact in the engineering design process. Vygotsky described the importance of culture and social interaction in the learning process. Previously, I had interpreted this theory to primarily be an interplay between teaching and learning - a balance between instructors and students. I also believed it prioritized being uncomfortable in order to fully be in the zone of proximal development. However, my team showed me that it is still effective when in a comfortable environment due to the value of peer discussion in building understanding. As shown in the feedback in figures three and four, we helped each other develop as people and engineers. I significantly felt growth in understanding because I had a safe space to ask questions and receive helpful feedback. This also widened my understanding of the engineering design process to have importance on the team dynamics. Although there should be procedures in place to keep work flow and ensure deadlines are met, creating a supportive space leads to better products. Having to navigate the process is inherently uncomfortable, but a safe space helps and can make it fun.

For the Future: Dynamic and Deadlines

I will be intentional in creating a safe and collaborative team dynamic in the future. This will allow growth and productivity. I also want to ensure we have agenda's and internal deadlines to ensure projects get done without sacrificing the fun of working with friends.

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Figure 1:



Figure 2:

As a group, we developed a really *great team dynamic* throughout both the design brief and alpha release assignments. We were able to create a really *positive team energy*. Our meetings were not only productive, but also *enjoyable, and this really allowed us to grow as a team and deliver great work*. We were able to foster an environment where everyone felt welcome and comfortable to share their ideas, no matter how crazy (for example, teleporting snacks!). These ideas actually allowed us to *push our creativity* and test how far we can go when creating these $n + 1$ concepts. Each team member was willing to take on work and contribute to the team's goals. We were able to communicate effectively and everyone was prompt in checking the group chat. Next time, one area that we could continue to *improve on as a team is to establish clear internal deadlines and work hard to ensure that we meet them*. Through the design brief especially, we learned that our team requires time to review and edit documents thoroughly before submission, as it allows us to refine our ideas and ensure we have met the criteria. By keeping this in mind for future assignments, we can create even better pieces of work that represent our team values. Overall, we worked really well together as a team and I am super happy with all that we have accomplished!

We overall *work extremely well as a team*. We are open to ideas, have clear expectations and values, and are willing to help and support each other wherever needed. Everyone is willing to meet and contribute what is necessary for success. We get along as people very well and are all becoming friends. I particularly admire how well we adapt to changing circumstances. When things go wrong we keep pushing and still get assignments done. We *collaborate and work effectively*, accounting for the success in our performance thus far. I believe we can continue to improve on these aspects of our team and *push each other to grow*. Sometimes, our bonds can result in us getting *sidetracked on personal conversations during meetings*. Although this *improves our team dynamics* and helps us form better connections with each other, it can *impact the duration of time needed to complete assignments*. This inefficiency may make things feel *draining* and can make tasks feel like more of an undertaking than they really are, making it harder to get started on new things. We should continue to focus on being more focused on the assignment during work periods. We can still have fun personal

Figure 3:

I find that we **worked very well as a team**, and I felt very **comfortable sharing my ideas**. This was a psychologically safe space, and being a member of this team **helped me grow in both the engineering design aspect and as a person overall** in terms of communication and teamwork abilities. I appreciate how **supportive and positive** our team always was, and someone was always present to motivate others. During the last few Alpha Release meetings we held, I admired how dedicated everyone was, and I feel inspired by all my teammates' open-minded attitudes when approaching problems. Moreover, I believe our team was very **helpful to each other**. If anyone ever struggled with any aspect of research or with making prototypes, someone else would always be present to help. This was something I appreciated a lot, as it boosted my confidence in our team's reliability. **Something I think our team needs to work on in the future is having clear goals for each meeting and not getting sidetracked**. Generally, we tend to **stray into unrelated conversations** as the meeting goes on, and we spend **longer completing the work than necessary**. This was great for team bonding; however, I find it negatively **impacted our efficiency**. For future meetings, I believe we can write out an agenda and make sure to complete all the tasks on the list within a certain time period.

As a team, we **work very well together in terms of idea generation, communication, and coordination**. Multiple members consistently step up to provide leadership: planning when and where we will meet, hosting and facilitating online meetings, and keeping track of what everyone is working on so that tasks don't fall through the cracks. During studios, we make **strong use of diverging tools, sketches, and clear formatting on paper and slides, which makes our concepts and prototypes easy to understand and visually organized**. Several teammates also contribute complex, creative ideas and think very critically about the weaknesses and limitations of

Figure 4:

- Great teamwork, everyone spoke equally and clearly. Optics of team showed good unity and cohesiveness.

Figure 5:

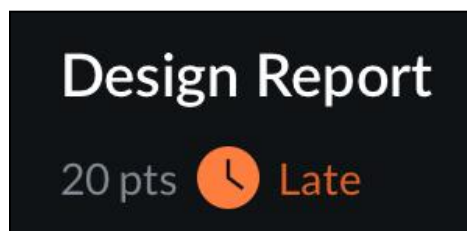


Figure 6.1:

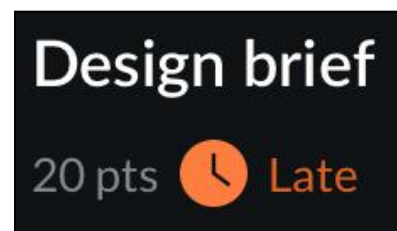


Figure 6.2: